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The illogic of being data-driven: reasserting control and restoring balance in our relationship with data and technology in football

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ABSTRACT

The ongoing advances and prevalence of technology in football is spurring a more prominent role for technology and data in driving decisions and practice. Sports science practitioners naturally play an integral part in the adoption of these technologies and are among the most enthusiastic proponents of a *data-driven* approach that promotes objective metrics and seeks to minimise or eliminate subjectivity. With the speed of these developments, there are a number of pitfalls that have been overlooked. In this commentary we describe the pervasive forces driving the adoption of technology solutions and critically examine the logic underpinning the present drive for metrics-based practice in football. In addition to highlighting some important gaps in what we are presently able to capture, we uncover some common flaws in how we implement new technology and use data. We propose that there is a fundamental need to reframe how we are seeking to employ data and more specifically make the necessary switch from being data-*driven* to data-*informed*. We propose some solutions to assist practitioners in being purposeful in their use of these tools and leverage the benefits of technology and data in a way that better supports decision-making and complements coaching practice.

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Introduction

Stakeholders in elite football are racing to harness advances in sports technology and analytics to capture and leverage data in the hope of deriving a competitive advantage (Cummins et al. 2013; Robertson 2020). The increasingly extensive application of technologies such as wearable sensors and analytical software in daily training environments has led many sport science and medicine practitioners to strive for a *data-driven* approach to decision-making, whereby key performance decisions are made 'by the numbers'.

In contrast to individual sports in which the performance of elite athletes between competitions is relatively stable (typical variation of ~1-4%) and predictable (Malcata and Hopkins 2014), the complexity and uncertainty involved in team sports such as football is much higher. Both outcomes and key performance indicators are subject to a host of additional factors, notably team dynamics (Higham et al. 2014). Therefore, effective decision-making within a competitive football environment is likely contingent upon much more than we can currently objectively capture and quantify.

Here, we outline the illogic of being *data-driven* in the context of football and present the rationale for why *data-informed* practice likely better supports our quest for a true competitive advantage.

What is driving us to be 'data-driven'?

The exponential acceleration of technology over recent years (Diamandis and Kotler 2020) has fuelled an arms race for sport science and medicine practitioners to own the latest and greatest tools to capture athlete performance and injury data. The availability, affordability, ease of use, and bold marketing claims accompanying these tools have led to their widespread adoption and implementation in elite football environments. The burgeoning data output from such hardware tools has spurred a parallel drive to acquire software tools to process, aggregate and visualize these numbers as metrics that provide simple and direct information to drive athlete performance decisions.

Simultaneously, there have been prominent examples in other sports of teams gaining a *Moneyball*-style competitive edge through the use of quantitative data (Lewis 2004). All of this has led to the prevailing belief that we should seek to be objective and driven by hard data, rather than being subjective and opinion-based in our judgement and decision-making. The basis for this contention is that objective sources of information are not subject to the same cognitive distortions and bias that plague our subjective reasoning.

So what is the problem?

Sport science and medicine practitioners within teams face constant pressure to be an early adopter of the latest technology solutions (Chambers et al. 2015), which can be problematic in multiple ways. The principal issue is that products may be adopted without practitioners being clear on what questions they are seeking to answer and what will move the performance needle within their team.

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We should acknowledge the industry influence on practice in the field: sports science staff in football are constantly inundated with promotional material from technology companies, with little regulation of marketing claims. Not only are early versions of technologies on the market often fraught with reliability issues but independent research frequently uncovers issues of validity regarding the outputs of technology that is in use. Examples include claims that wearable technology was able to provide the resolution to detect match activities (such as kicking and tackling) (Chambers et al. 2015) and accurately quantify energy expenditure (Gastin et al. 2018).

Due diligence is thus required before adopting technology solutions (in reference to the questions we seek to answer), otherwise we forfeit our ability to understand what is signal and what is merely noise or random variation. By extension, we perhaps too readily put our faith in the 'black box' and the tools used to acquire and process the data (Malone et al. 2017). Much like the methods and algorithms employed, the validity of the metrics derived has also escaped our scrutiny on occasion. An example that should serve as a cautionary tale is the widespread use of the *acute: chronic workload ratio* approach, whereby a conceptually and methodologically questionable metric was used to regulate training load (Impellizzeri et al. 2020).

Finally, as we rush to adopt technologies in the name of eliminating subjectivity, we overlook the fact that how we select and use these tools is itself subject to influences that are not necessarily rational. Indeed, a key reason why the acute: chronic workload ratio methodology was adopted so enthusiastically and ubiquitously among sport science and medicine practitioners was because it appealed to their respective confirmation biases; their effort gathering GPS data was worthwhile and injuries were due to training errors.

Beyond the dangers of misuse or overuse of data in decision-making, the introduction of any data or metrics has the potential to distract and even detract from the intended purpose. The act of measurement itself can influence perceptions, motivations and in turn behaviours (McGuigan 2017; Gamble 2020). When certain metrics are assigned importance, this compounds the behavioural drive towards the metric itself and away from the performance it was intended to support. This phenomenon is encapsulated by Goodhart's Law, which cautions: 'When a measure becomes a target, it ceases to be a good measure.' In football, players can become distracted from the original pursuit of performance in favour of proxy measures. *Velocity-based training* provides a good illustration of a current training practice that is prone to players chasing numbers rather than execution (Gamble 2020).

What does the data fail to capture?

The *data-driven* approach favours the simple and easy to quantify; whatever is measurable is given more weight when making decisions. While numbers alone may be useful for solving simple-system problems (e.g. pass-fail test), this approach is too reductionist for understanding human behaviour and performance. Complex systems – particularly complex adaptive biological systems with a mind attached – present much more of an intractable problem (Robertson and Joyce 2019; Gamble 2020).

Despite advances in technology, there is much within human behaviour and performance that we are still unable to capture through numbers. Inevitably, there is a need to fill in these blanks and connect the dots. A coach's skilled intuition and subjective assessment of their players thus remains invaluable and irreplaceable. For example, a recent study of elite athletes found that coach judgement independent of all athlete monitoring variables was the most sensitive predictor of performance (Crowcroft et al. 2020). Such findings reinforce the assertion that the best athlete monitoring tool is a coach who pays attention.

While data and metrics cannot adequately account for human behaviour, the objectivity they provide is nevertheless valuable. Humans cannot compete with technology when it comes to capacity for information processing, and we are prone to biases and influences that affect our judgement (Robertson and Joyce 2019). A shared decision-making process with objective quantitative data that complements subjective observations and qualitative assessment would harness the strengths of both 'man' and 'machine' (Gamble 2020; Robertson 2020). For instance, combining coach evaluations with data from multidimensional performance tests demonstrated superior predictive accuracy for talent selection in youth football (Sieghartsleitner et al. 2019). As the authors concluded, 'the combination of the subjective coaches' eye with scientific data may buffer the mutual weaknesses of these different approaches'.

How should we reframe the role of data?

To leverage data for competitive advantage, we must first break down the false dichotomy between objective and subjective (Fullagar et al. 2019). Instead of a choice between *data-driven* decision-making or reliance upon subjective professional judgement that excludes the use of data, we must take a more nuanced view. Reframing technology and data as tools that supplement subjective assessment as opposed to replacing their input allows for a deeper understanding of complex human behaviour and therefore better decisions. Through defining the complementary role of data, we also challenge the perception that data and metrics undermine or override the professional judgement of coaches, likely inducing greater respect from coaches for the value of what data can teach us. All in all, we should strive to become *data-informed* rather than *data-driven* in our practice.

Becoming *data-informed* starts with clarifying the key questions we are seeking to answer. Once defined, we can determine what data will serve that purpose and then select our tools accordingly, as opposed to investing in technology and hunting for applications. As was recently advocated for player monitoring technology such as GPS, we must be prepared to undertake in-house verification of validity and reliability prior to use (Malone et al. 2019). Specifically, we need to establish whether devices capture what they purport to (and with what precision). Fundamentally, we must also determine the repeatability of the measure – otherwise, all we have is an expensive random number generator. When we have multiple units – such as GPS and accelerometer devices – we must further determine the agreement

between units as this cannot be assumed (Nicolella et al. 2018). Clearly we need to verify inter-device reliability and accuracy in order to have a sound basis to compare scores between players or between days if using different units.

These considerations are particularly pertinent when considering tools that derive metrics from raw data, given the potential for compounding errors (Lolli et al. 2019). Simultaneously, practitioners should also seek to understand the smallest worthwhile change in their data for their specific context (Hopkins et al. 2009), such that meaningful changes over time can be distinguished from random variation. Issues of sensitivity with respect to GPS- and accelerometer-derived peak external load metrics have been raised with rugby football for example (Howe et al. 2020).

Beyond exercising due diligence in how quantitative data are collected, analysed and interpreted, it is important that we are selective in how the information derived is employed with players and coaches. We must be intentional about what is shared, when and with whom. We need to be pragmatic about the limited bandwidth that players and coaches have for metrics. Surveys of high-level football indicate that over 50 different external load variables are presently used by the performance staff at different teams (Akenhead and Nassis 2016). Indeed 'too much information' is specifically cited by football coaches as a barrier to making use of the input provided by their sports science staff (Nosek et al. 2020).

It is equally critical that we seek to prevent the actualization of Goodhart's Law. Conversely, with respect to the potential for coach- or player-facing data to influence behaviour, perhaps we could also consider leveraging this to our advantage by using selective measures to reinforce desired behaviours. For example, many mobile education apps present users with daily-use streaks to reward their continued effort towards long-term goals.

We must also be clear about the limitations and uncertainty in our data, in relation to the questions we are answering. For simple problems, we may be confident enough to make automatic decisions using data. For more complex problems of human behaviour, we must realise data are not sentient and we cannot expect them to do the thinking for us. Being *data-informed* means knowing when data can replace conversations and spare cognitive load, but otherwise using data to provide additional insights that guide and better inform decision-making.

Finally, *data-informed* practice encompasses efforts to develop our professional judgment and decision-making through the use of data. Within the *professional judgement and decision-making* framework proposed by Abraham and Collins (Abraham and Collins 2011), data can create objective feedback loops to calibrate, evaluate and review our decisions. For instance, a useful practice is to assign numbers to our confidence level when making decisions and recording them for later review (Tetlock and Gardner 2016). This is one example of how data can provide a *decision support system* to identify factors associated with better decisions and highlight flaws in our reasoning (Schelling and Robertson 2020).

Concluding remarks

Whilst the recent advances in football technology and analytics might often seem like a silver bullet, they do not remove the role of humans in the decision-making process. The abundance of tools and data at our disposal should place greater onus on our critical thinking, not less. Ultimately, our rivals are likely using the same tools, so our competitive advantage surely lies in uring out how best to find the right signals in a world full of noise. Indiscriminate adoption of technology and mindless application of metrics in decision-making can be expensive, time-consuming and potentially detrimental to the outcome. We must be selective and clearly define the purpose that our data and metrics serve. We should also be mindful of the potential pitfalls and paradoxical second-order effects that data create so we can take steps to mitigate unintended consequences.

Striving to be *data-driven* is not a logical aspiration given we are dealing with complex adaptive systems that also have a mind attached. The quest to remove all subjectivity is similarly not only flawed but also quite futile. To paraphrase Einstein, not everything we measure matters and not everything that matters is measurable (at least at present). Subjectivity is inescapable given the inherent complexity and uncertainty of human performance. In football, we are inevitably working with incomplete information. Subjective reasoning honed through experience, education, and expertise thus remains an integral part of professional judgement and decision-making in sport (Collins and Collins 2015). The role of technology and data in general is to inform, facilitate and elevate our decision-making capabilities. If we begin with this proposition of being *data-informed* rather than *data-driven* we can finally move past the tension between objective versus subjective and turn our attention to blending these approaches in a way that is mutually complementary. After all, as Hans Rosling said, '*The world cannot be understood without numbers, and it cannot be understood with numbers alone*' (Rosling 2019).

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